Abstract template for the Advanced School in Applied Machine Learning: Uncertainty and Robustness of DL Models for Science — (smr 4228)

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This document is the template for contributed abstracts posters and gives the guidelines for preparing and submitting abstracts for the *Advanced School in Applied Machine Learning: Uncertainty and Robustness of DL Models for Science — (smr 4228)* to be held in-person from 23/07/2026 to 31/07/2026.

We present a robust deep learning framework for antimalarial drug discovery that addresses uncertainty quantification and model reliability through variational autoencoders (VAEs), consensus validation, and systematic performance assessment. Our approach demonstrates how uncertainty-aware deep learning can enhance scientific discovery in high-stakes pharmaceutical applications where model reliability is critical.

Uncertainty Quantification in VAE Models: We trained variational autoencoders with 32D and 64D latent spaces to learn probabilistic molecular representations from 65,856 compounds. The VAE architecture incorporates explicit uncertainty through KL divergence regularization (12.86 for 64D model), enabling principled uncertainty estimation in chemical space embeddings. Reconstruction accuracy reached 93.23% with validation loss convergence demonstrating model stability. We quantified epistemic uncertainty through latent space variance and aleatoric uncertainty through reconstruction confidence, providing uncertainty bounds for molecular similarity predictions.

Robustness Through Consensus Validation: To address the inherent uncertainty in molecular docking predictions, we developed a dual-filter consensus strategy combining AutoDock Vina (physics-based) and DiffDock (deep learning) methods. This orthogonal validation approach reduces false positive rates by 12-45% across four *Plasmodium falciparum* targets, with correlation analysis ($r = 0.1$-0.4) confirming complementary information content. The consensus framework provides robust predictions by requiring agreement between fundamentally different computational approaches, effectively quantifying prediction uncertainty through method disagreement.

Scalable Uncertainty-Aware Pipeline: Our framework processes large-scale chemical libraries (65,856 compounds) with computational efficiency while maintaining uncertainty tracking throughout the pipeline. Clustering quality metrics (Silhouette: 0.229, Calinski-Harabasz: 1501) provide confidence estimates for chemical space partitioning. Multi-parameter optimization balances prediction uncertainty across five objectives (docking, ADMET, drug-likeness, synthesis, activity) with explicit confidence intervals for each component.

Validation and Robustness Assessment: We validated model robustness through: (1) Cross-validation of VAE latent dimensions (32D vs 64D comparison), (2) Clustering algorithm comparison (KMeans, BIRCH, Jarvis-Patrick), (3) External validation using DEKOIS 2.0 decoy sets and MMV Malaria Box benchmarks, (4) Sensitivity analysis of hyperparameters and thresholds. Perfect ROC-AUC (1.000) in retrospective analysis demonstrates model reliability, while enrichment factors (EF5% = 1.78) provide confidence in hit prioritization.

Scientific Impact: The uncertainty-aware framework identified 19,913 synthesizable leads with quantified confidence scores, including > 70 polypharmacological candidates. Explicit uncertainty quantification enables risk-stratified experimental validation, optimizing resource allocation in drug discovery pipelines. The approach demonstrates how robust deep learning can accelerate scientific discovery while maintaining reliability standards essential for pharmaceutical applications.

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